



Evolution of LiverNet 2.x: Architectures for automated liver cancer grade classification from H&E stained liver histopathological images

Amit Kumar Chanchal¹ · Shyam Lal¹  · Dipanshu Barnwal¹ · Prince Sinha¹ · Shrikant Arvavasu¹ · Jyoti Kini^{2,3}

Received: 20 October 2022 / Revised: 11 March 2023 / Accepted: 27 March 2023 /

Published online: 15 May 2023

© The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2023

Abstract

Recently, the automation of disease identification has been quite popular in the field of medical diagnosis. The rise of Convolutional Neural Networks (CNNs) for training and generalizing medical image data has proven to be quite efficient in detecting and identifying the types and sub-types of various diseases. Since the classification of large datasets of Hematoxylin & Eosin (H&E) stained histopathology images by experts can be expensive and time-consuming, automated processes using deep learning have been encouraged for the past decade. This paper introduces LiverNet 2.x model by modifying the previously encountered LiverNet architecture. The proposed model uses two different improvements of the Atrous Spatial Pyramid Pooling (ASPP) block to extract the clinically defined features of hepatocellular carcinoma (HCC) from liver histopathology images. LiverNet 2.0 uses a modified form of ASPP block known as DenseASPP, where all the atrous convolution outputs are densely connected. Whereas LiverNet 2.1 uses fewer concatenations while maintaining a large receptive field by stacking the dilated convolutional blocks in a tree-like fashion. This paper also discusses the trade-off between LiverNet 2.0 and LiverNet 2.1 in terms of accuracy and computational complexity. All comparison model and the proposed model is trained and tested on the patches of two different histopathological datasets. The experimental results show that the proposed model performs better compared to reference models. For the KMC Liver dataset, LiverNet 2.0 and LiverNet 2.1 achieved an accuracy of 97.50% and 97.14% respectively. Accuracy of 94.37% and 97.14% for the TCGA Liver dataset are achieved.

Keywords Atrous convolutions · Convolutional neural networks · Deep learning · Dense connections · Histopathology images · Medical image classification

✉ Shyam Lal
shyam.mtec@gmail.com

1 Introduction

According to Calderaro et al. [7], liver cancer is the fifth in the list of highly frequent cancers worldwide and also the fourth largest cause of cancer-related deaths. Liver cancer is caused due to abnormal changes in the deoxyribonucleic acid (DNA) of liver cells. This change leads to the rapid growth of abnormal cells and results in tumors. Late diagnosis is the primary reason for the high rate of mortality by liver cancer, which significantly diminishes the chances of survival of the patients. Hepatocellular carcinoma (HCC) is one of the most common types of liver cancer that develops in the liver cells, especially in African and Asian countries. The monitoring of HCC is essential for the early detection of liver cancer and thus would improve overall survival [44]. The conventional methods involve the manual examination of microscopic tissues of histopathology slides. In the manual examination, the cancerous cells are stained with hematoxylin and eosin stains (H&E stain). The purple-blue stain on cell nuclei is due to hematoxylin and the pink stain on the cytoplasm and extracellular matrix is due to eosin. Specific features like the size and the stained color of nuclei, distribution of the nuclei, and the shade of the cytoplasm due to the stain are observed by the pathologists. Slides of malignant (cancerous) tissues can be distinguished from benign tissues if some specific indicators show up. Abnormalities like increased distribution and decentralization of nuclei and a darker stained cytoplasm become more intense in the later stages of cancer. Considering these indicators, a pathologist can comment on the type or stage of cancer. To estimate the sub-type of liver HCC tumor, cancer grading is done based on the morphological features of the tumor. A grading score given to the tumor is based on the cellular differentiation of the tissues. Grading of tissues highly depends on the comparison of normal tissue with cancerous tissue discussed by Travis et al. [39]. The difficulties arising from conventional methods are often mitigated by using computer-aided detection (CAD) systems developed using image processing techniques. Automated processing of histopathology images can be beneficial as evaluations by experts can be costly and time-consuming. Earlier, self-supervised techniques were used for histopathology image analysis and later, semi-supervised methods were developed for effective classification tasks discussed in [34]. Methods such as fuzzy detection algorithms and feature-oriented dictionary learning [40] have also been proposed in the past few years. These methods do have their limitations, where the sparse coefficient of input data is generated using sampling techniques. Sparse coefficients are interpreted as features and the level of sparsity depends on data representation at each stage. Here errors accumulate in multiple stages of the process that causes to focus only on major variances in data whereas minor details which are crucial for discrimination are also equally important. Few innovative ideas of the convolutional neural network on different image modalities have been reported in [3, 33] in which transfer learning, k-fold analysis, and fusion of selected features have been performed. Recent studies on liver histopathology images [9, 12, 28, 35] apply computationally efficient block like SENet, residual learning, and inception module to extract representative features. The work in [43] uses a support vector machine to build a classification model. The main objective of this paper is to design an efficient automated system for the classification of HCC into four stages of cancer. The various steps involved in the proposed method include pre-processing, feature extraction, and classification. The proposed model is intended to show how deep learning can be applied effectively to differentiate liver histopathology images into four categories, namely Grade-0, Grade-1, Grade-2, and Grade-3. The deep learning-based feature extraction approach has gained significant popularity and attention in digital pathology due to its promising results and high prediction rate. Although CNN-based deep learning methods have proven to be better than the conventional method of medical image

processing for extraction of the crucial features from the histopathology images, it has been challenging to grade the cancers because of these reasons.

1. The clinical capability of the deep learning model is affected due to the wide variability of H&E stain across the entire liver histopathology images.
2. Conventional algorithm prediction process is affected due to highly imbalanced classes in the liver histopathology images.
3. Labelling different liver HCC cancer grades from H&E stained digital histopathology data requires a highly skilled pathologist.
4. To maximize the classification accuracy of the method, emerging of more and more developments resulted in a heavy computational load.

There is a need for a fully automated deep-learning solution to overcome the challenges in digital pathology. The final goal of using Artificial Intelligence (AI) in healthcare is to design a fast and precise cancer detection system. The proposed method provides a computationally efficient and fully automated deep learning framework, which is one step ahead compared to existing solutions. Our approach is generalized and effectively works on two different histopathology datasets. The contributions of our research work are as follows.

1. A LiverNet 2.x has been introduced for automated Liver cancer Grade classification from H&E-stained liver histopathology images.
2. Experimental results of the proposed LiverNet 2.x architecture have demonstrated two histopathology datasets in terms of quality of classification and computational efficiency.

2 Related work

There has been a rising growth of AI and machine learning, especially deep learning architectures in science, engineering, and technology. The field of medical imaging is one of them, where accurate diagnosis and medical data processing techniques [32] have advanced steadily in recent years. The process of automating histopathology classification has been experimented with by various researchers. In [37], authors introduce a fuzzy decision logic using fuzzy decision trees to measure the degree of human epidermal growth factor receptor 2 (HER 2/neu) receptor's over-expression. In [4], authors compare the performance of several features like local binary patterns (LBPs), histogram of gradients (HOGs), and deep features from a pre-trained deep neural network. Convolutional Neural Networks (CNNs) proved to be quite effective and robust in extracting features from spatial data. Several state-of-the-art architectures like AlexNet [24], ResNet [18] and GoogLeNet [36] have been introduced in the past decade with ground-breaking results in datasets like ImageNet, COCO, and BreKHis. Though they have shown superior performance in their precise and reliable feature extraction, they suffer from a large number of trainable parameters, thus making it difficult for them to be deployed for healthcare applications. However, it is interesting to note the architectural advances they propose, which can be exploited along with some other concepts like attention, deep supervision, etc. to come up with various less-complex deep learning architectures for achieving comparable experimental results. Some of those architectures related classification of histopathology images are discussed here. BreastNet [38] was among the first to introduce attention modules for the breast cancer classification task. This improvement over the other state-of-the-art architectures solves the problem of precisely classifying tumors into several classes (depending on the grade of the

cells) rather than just two (benign and malignant). Their approach relied on the channel attention and spatial attention modules used. The general structure of the given model is based on residual architecture. This architecture addresses the problem of image features of varying scales. The attention module used is responsible for the selection of key regions of the images. After max-pooling at three different layers in the model, the expansion is done using the bilinear upsampling layer instead of transpose convolutions to reduce the number of parameters. The model performed better than the architectures discussed earlier and this lays a foundation for the development of the LiverNet architecture. LiverNet [1] deals with the multi-class classification of liver HCC histopathology images among various grades of liver HCC tumors. It not only shows a greater performance but does so with lesser computational efficiency. This efficiency in the improved performance is attributed to the addition of atrous spatial pyramid pooling (ASPP) block to the BreastNet architecture just before upsampling. This atrous spatial pyramid pooling (ASPP) block [11] helps in detecting multi-scale features with the increased receptive field while keeping the number of parameters involved as same. The work by Hameed et al. [15] leverages the strength of inception and residual connections, and the applied method is computationally efficient using depth-wise separable convolution. The experiment in [2], where sixteen different networks pre-trained on ImageNet weights have been studied, indicated that the idea of transfer learning in the case of breast cancer histopathology images is effective. For transferred features of ImageNet data, the Inception-ResNet V2 model performs better compared to the state-of-the-art deep learning models. To classify and identify the four major subtypes of breast histopathology images Nguyen et al. [30] experimented with different versions of the EfficientNet model. The obtained classification accuracy is better than state-of-the-art deep learning models. Multiple image modalities of four different groups as computed tomography (CT) of the brain, CT images of the abdomen, magnetic resonance imaging (MRI) of the spine, and MRI images of the brain are converted to joint photographic experts group (JPEG) format and classified using CNN framework [13]. A hybridized CNN model using pre-trained ResNet50 and an improved marine predators algorithm (IMPA), which is an improved optimization algorithm. The proposed architecture called IMPA-ResNet50 [19] is effective in two mammographic datasets, called mammographic image analysis society (MIAS) and curated breast imaging subset of DDSM (CBIS-DDSM) datasets. The proposed AlexNet-BC [27] used some additional fully connected (FC) layers to the original AlexNet model to construct four different types of networks. The mentioned results show that an additional FC layer to the original AlexNet is able to increase the nonlinearity of the network and improves accuracy. For binary classification of ultrasound and histopathology images, the proposed model in [5] used customized AlexNet. The modified AlexNet empowered with transfer learning is efficient for different image modalities. In [16], Gray-level co-occurrence matrix (GLCM) features and the extracted deep features by the DensNet201 are concatenated and finally support vector machine (SVM) is used as a classifier for the classification of breast cancer histopathological tissues. The work in [12] uses a pre-trained Inception V3 network on ImageNet weights. Some top layers are added to fine-tune the parameters and generate the probabilistic distributions for each class of liver histopathology images. Sun et al. [35] extracted patch-level features using transfer learning and then the image label feature descriptor is formed by combining patch-level features.

The proposed deep learning classification method in [35], where liver histopathology images are classified into cancerous and noncancerous using only global labels. Chen et al. [9] applied a lightweight CNN module called squeeze and excitation (SE) for the classification of liver histopathology images. This study demonstrated different types of attention mechanisms in deep learning and the effectiveness of attention visualization in the feature

extraction of medical images. The study in [31] used transfer learning, a combination of efficient CNN networks like (DeepLapV3 + ResNet-50), (VGG-16 + U-Net + ResNet-50V2 + LSTM) to diagnose CT images of liver cancer. A concise summary of state-of-the-art deep learning techniques used for the classification of medical images which includes application, method, deep learning tools used, performance measures, and limitations are presented in Table 1. In this work, a LiverNet 2.x has been introduced to perform automated liver cancer grade classification from H&E-stained liver histopathology images. Further, LiverNet 2.0 has been introduced to perform automated Liver cancer grade classification from H&E-stained liver histopathology images with the objective to provide higher classification results. LiverNet 2.1 is the variant of LiveNet 2.0 and it has been introduced to perform automated liver cancer grade classification from H&E-stained liver histopathology images with the objective to provide better classification results with higher computationally load.

3 Proposed methodology

The main contribution of this paper lies in the modifications made to the LiverNet architecture [1]. The improvements mainly deal with the modification of Atrous Spatial Pyramid Pooling (ASPP) block. Like the ASPP block, these blocks are also implemented using separable convolutions, so as to have the number of parameters as minimum as possible, while making the architecture efficient. The extent to which the less number of parameters prove to be critical in the efficiency of a model is discussed in further sections. We highlight the features of the proposed architectures LiverNet 2.0 and LiverNet 2.1, and then proceed with their tradeoffs. The workflow of the proposed work has been pipelined as shown in Fig. 1. The improved architectures are discussed in the following section in detail.

3.1 LiverNet 2.0

The central idea of LiverNet, which is an improvement over BreastNet, was the inclusion of atrous convolutions. Atrous Convolutions were introduced to learn features with large receptive fields, while not sacrificing the spatial resolution. The idea of the ASPP block was to combine the use of features at various scales and receptive areas. Though ASPP blocks were able to learn feature representations from different scales, Yang et al. [45] argued that the feature resolution in the scale-axis was not dense enough, thus proposing dense connections to the ASPP block, which was quite similar to the dense connections used in the DenseNet [20]. We have analyzed the motivation behind the dense connections and found out that this improvement can be incorporated into the LiverNet architecture as well. The blocks used in the LiverNet architecture, shown in Fig. 2, are discussed below.

1. Convolutional and linear blocks These blocks involve a 2D convolutional layer and linear layer (also known as a fully-connected layer) respectively, followed by batch normalization [21] and ReLU activation function. Convolution is an operation of two functions. For two-dimensional images, we usually deal with a two-dimensional kernel. Two-dimensional images and kernels are matrices, which have some numbers. The mathematical expression describing the 2D convolution operation is presented in (1), where $C(x, y)$ is the convoluted signal, K is the kernel, I input image, n, m are kernel indexes.

$$C(x, y) = K * I = \sum_{n,m} I(x - n, y - m)K(n, m) \quad (1)$$

Table 1 Summary of state-of-the-art deep learning techniques used for classification of medical images

References/Application	Method	Tools used	Performance Criteria	Limitations
Hao et al. [16]. • Breast cancer histopathological images classification.	• DenseNet201 is used as a base model and extracted features are fused with the three-channel GLCM features. • Support vector machine (SVM) is used for classification.	• Nvidia GeForce GTX 1080Ti GPU, Stochastic gradient descent (SGD) with momentum factor of 0.9 and initial learning rate was set to 0.0001. • Cross-entropy loss function is used.	• Image-level recognition accuracy is 96.75%, 95.21%, 96.57%, and 93.15% for different magnifications. • Patient-level recognition accuracy is 96.33%, 95.26%, 96.09%, and 92.99% for different magnifications.	• This study does not include a multi-class classification of breast cancer histopathological images.
Hameed et al. [15]. • Multi-class classification of breast cancer histopathology images.	• Inception modules in conjunction with residual connections are used. • Global pooling at six different layers and these multilevel features are then concatenated horizontally.	• Python version 3.8.5 and TensorFlow 2.4.1 • Nvidia GeForce GTX 2070 graphical processing units. • Adam optimizer.	• Accuracy = 98%, Kapa Score = 0.995, AUC-ROC Score = 0.998, and Mean AUC-PR = 0.995.	• There is lack of comparison with state-of-the art methods.
Abbasniya et al. [2]. • Classification of breast tumors based on histopathology images	• Inception-ResNet-v2 is used where CNN model has both residual and inception networks. • A transfer learning approach. • LightGBM, XGBoost, and CatBoost classifier.	• Keras implementation of the pre-trained networks is used. • The Google-Colaboratory free environment. (12 GB RAM, 78 GB disk, and cloud GPU) • Grad-CAM.	• Accuracy of 96.83%, 95.84%, 97.02%, and 96.16% on the 40X, 100X, 200X, and 400X magnification factors respectively.	• Only the pre-trained network is focused.
Nguyen et al. [30]. • Histopathological imaging classification of breast tissue obtained from ICIAR 2018 grand challenge.	• EfficientNet model based on scaling principle is used to classify and identify the histopathological images of breast tissue. • Uniform scaling for all three dimensions with a fixed ratio is performed.	• Python scikit-learn module is used. • Colab Pro environment. • Learning rate scheduler.	• 98% accuracy on the training set and 93% on the test set.	• A restricted quantity of images is used to train the model. The obtained results are not satisfactory.

Table 1 (continued)

References/Application	Method	Tools used	Performance Criteria	Limitations
Houssein et al. [19]. • Breast cancer detection and classification using two mammographic datasets.	• ResNet50 model is hybridized with the improved marine predators algorithm (IMPA-algorithm). • Transfer learning is used.	• Keras ImageDataGenerator. • Hybridized CNN. • Google Colaboratory platform. • Python 3 with Keras. • Stochastic gradient descent.	• Accuracy = 98.32%, Sensitivity = 98.56%, and Specificity = 98.68% on CBIS-DDSM dataset. • Accuracy = 98.88%, Sensitivity = 97.61%, and Specificity = 98.40% on MIAS dataset.	• Computationally expensive. Results are limited to a specific dataset.
Liu et al. [27]. • Breast cancer classification in the pathology images.	• A modified form of AlexNet, which uses different variations of fully-connected layers. • Using the pre-trained weight of ImageNet dataset. • A loss function, where a penalty term is added to the original cross-entropy.	• Intel i7-9700 3.0GHz CPU. • NVIDIA Quadro RTX 4000 GPU. • PyCharm IDE with Python 3.6.4 • Tensorflow 1.8.0 and Keras 2.2.4	• Average accuracy is $98.15 \pm 0.9\%$, $97.71 \pm 1.9\%$, $97.96 \pm 0.7\%$, $98.48 \pm 1.1\%$, for $40 \times, 100 \times, 200 \times, 400 \times$ respectively.	• Small scale dataset, the new loss function is not a generalized loss function.
Arooj et al. [5]. • Breast cancer detection and classification using ultrasound and histopathology images.	• Used three different datasets of breast cancer and evaluated by customizing the model AlexNet. • Transfer learning is used.	• MATLAB 2020a. • Stochastic gradient descent with momentum.	• Accuracy = 99.4%, 96.70%, 99.10%, 100% for four different datasets.	• Performance is not compared with state-of-the-art methods.
Chen C et al. [9]. • Classification of multi-differentiated liver cancer.	• Squeeze and Excitation is used. • Feature maps of different convolution kernel sizes are fused through the attention mechanism.	• IntelCore I5 9600KF CPU, NVIDIA GTX 1070 GPU. • TensorFlow V2.1.0, Keras V2.3.1. • Python V3.7.0. • Sci-kit image V0.16.2, Sci-kit learn V0.22	• Poorly differentiated (Acc)= 98.41 % • Moderate differentiation (Acc)= 92.59% • Well differentiated (Acc)= 95.30%.	• Dataset used for the experiments was small and imbalanced.
Othman E et al. [31]. • Automatic detection of liver cancer from CT images.	• Tested multiple combinations of pre-trained models. • Integrated the outputs of the last layer of multiple networks.	• TensorFlow with the Keras library. • Python latest version. • ReLu activation.	• Accuracy=0.995 • Precision=0.864 • Recall=0.979	• The architecture contains a massive number of weight parameters, resulting in big model size and inference time.

Table 1 (continued)

References/Application	Method	Tools used	Performance Criteria	Limitations
Aaresh et al. [1]. • Multi-class classification of Liver histopathology images.	• Incorporation of attention block, residual block, and Atrous spatial pyramid pooling (ASPP) block in deep learning.	• Nvidia Tesla K80 GPU, L2-regularization, Adam optimization. • Python and the Keras deep-learning library.	• KMC Liver Dataset = F1 Score=0.9093, IoU=0.836, Accuracy=0.9093. • TCGA Liver Dataset = F1 Score=0.9772, IoU=0.9561, Accuracy=0.9772.	• The used dataset is restricted in terms of size and variance.
	• The CNN architecture consists of multiple layers of Convolution, Batch normalization, ReLu, Max-pooling, and fully connected layers. • Classify four different groups of medical image modalities.	• Matlab 2020a. • Stochastic gradient descent with momentum. • Standard MacBook Pro computer. • Intel Core i7 2.7 GHz CPU, 16G main memory. • Intel HD Graphics 530 1536 MB.	• Accuracy greater than 99.5% • F1 Score greater than 99% .	• For multiple image modalities, the reported results are not generalized.
Togacar et al. [38]. • Eight class and two class classifications of breast histopathology images.	• Residual architecture built on attention modules. • Flip, shift, brightness change, and rotation are used as data augmentation techniques to train the model. • Designed CNN Model.	• NVIDIA GeForce GTX 1070 - 8 GB graphics card. • Adam and Stochastic Gradient Descent (SGD) optimization. • Python and Matlab.	• Accuracy = 98.80%, Sensitivity = 98.35%, Precision = 98.84%, and F1 Score = 98.59%.	• Lack of diverse dataset.
	• Inception V3 is used as a base model. • Used pre-trained model offered by TensorFlow. • The model was externally validated by an independent data cohort.	• The model training based on an easy-to-use platform called EASY DL. • Uses PaddlePaddle deep learning framework V3.0 created by Baidu Brain AI technology.	• For benign and malignant classification: Acc= 0.96, Pr= 0.945, Re= 1, F1=0.971 and MCC= 0.912. • For well, moderate, and poor tumor differentiation: Acc= 0.896, Pr= 0.879, Re= 0.771, F1=0.820 and MCC= 0.738.	• Dataset is small and results are limited to a specific dataset.

Table 1 (continued)

References/Application	Method	Tools used	Performance Criteria	Limitations
Sun et al. [35]. • Classification of Liver Cancer Histopathology Images.	• Transfer learning is used to obtain the patch-level features. • A patch-based method that divides the histopathological image into smaller patches.	• TensorFlow 1.0.0 • L2-regularization and dropout were used. • Adam optimizer, batch size= 24, and learning rate= 0.001	• Case-1, For K=10, F1=0.86, Acc=0.86 • Case-2, For K=100, F1=0.99, Acc=0.986.	• Weakly supervised learning, use of only global labels, and semi-supervised learning are unfavorable factors.
Kim et al. [22]. • Detection of the left ventricle in echocardiographic images.	• The proposed SegAN architecture is composed of segmentor and critic which performs as a generative adversarial network model. • Uses a multi-scale feature loss function	• Nvidia TESLA K80 GPU and NVIDIA GeForce GTX 1050 GPU, Adam optimizer with a learning rate of 0.001. • Python and TensorFlow.	• Model-1 Dice = 0.858± 0.066, Precision = 0.852± 0.089, HD = 6.43 ± 1.40 • Model-2 Dice = 0.855 ± 0.066, Precision = 0.852 ± 0.092, HD = 6.28 ± 1.119.	• Insufficient datasets, Focused on previously trained weights.

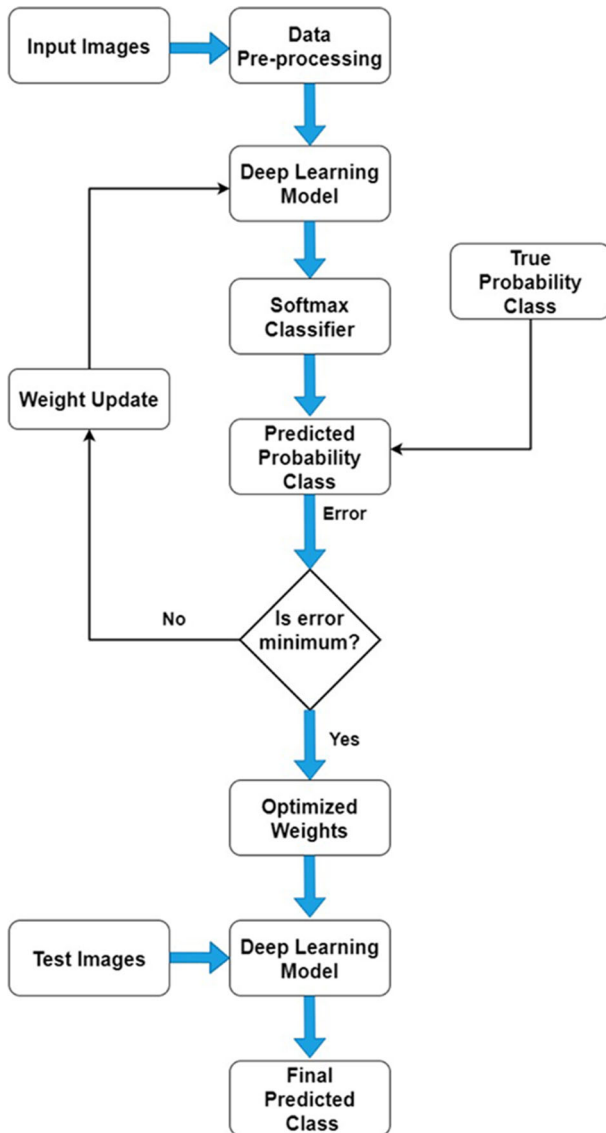


Fig. 1 Classification pipeline of proposed deep learning model

A fully connected (FC) layer is a very important layer when it comes to classifying an image into a label for classification problem perspective. In the FC layer every input node is connected to every output node by a weight. It serves the purpose of doing the actual classification. Without this layer, a traditional convolutional neural network (CNN) would be unable to split out the predicted classes. An image is composed of smaller details of features. CNN leverages this fact and utilizes its various layer for analyzing each feature in isolation, thereby informing a decision about an image as a whole. FC layer in CNN is the one that takes the end result of the convolution or pooling via the flattening layer and

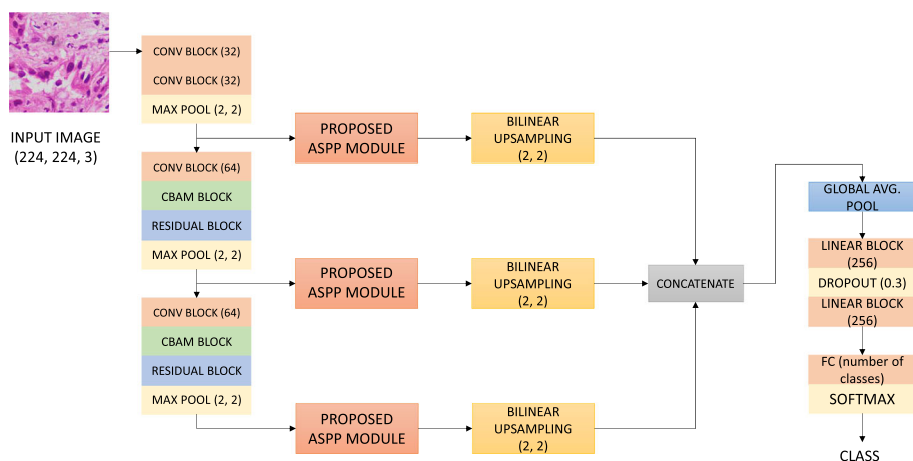


Fig. 2 LiverNet 2.x architecture

reaches the classification decision. So applying all the layers like convolution, ReLU, batch normalization, dropout, and flattening, the feature map is then passed through the FC layer. This FC layer combines the features into more attributes, which better predict the classes.

2. Residual block The residual block was inspired by the ResNet architecture [18], which is an influential work in deep learning and makes an interesting observation by adding the original input to the output of the convolutional block. This block consists of two convolutional layers and avoids the problem of vanishing or exploding gradients by using the concept of skip connection. Instead of convolution layers one after another which involves lots of parameters in the form of weights, they have proposed a skip connection shown in Fig. 3. These weight layers simply learn whatever makes the output different from this input. If $F(X)$ tends towards zero function then the output becomes an identity function. The default function of this network is the identity function.

3. CBAM block The convolutional block attention module (CBAM) is an effective attention module for CNNs, as the name suggests [42] and these modules are helpful for concentrating the essential features. It is responsible for obtaining the channel and spatial attention maps for an input feature map. This module consists of a channel-attention block and a spatial-attention block.

Let's consider a feature map¹ $F \in \mathbb{R}^{C \times H \times W}$, which acts as an input to the CBAM. The function of the CBAM is to provide channel and spatial-wise attention. The attention modules (considering channel-first mode) can be expressed mathematically using the functions $f_c : \mathbb{R}^{C \times H \times W} \rightarrow \mathbb{R}^{C \times 1 \times 1}$ and $f_s : \mathbb{R}^{C \times H \times W} \rightarrow \mathbb{R}^{1 \times H \times W}$. The CBAM thus implemented is expressed in (2), and (3).

$$F_1 = f_c(F) \cdot F \quad (2)$$

$$F_2 = f_s(F_1) \cdot F_1 \quad (3)$$

Here, ' $\cdot$ ' represents the element-wise multiplication and F_2 is the final output of the CBAM block. In the deep learning paradigm, the functions f_c and f_s are learned using

¹Note: We have used channel-first notation for the images and the corresponding tensors.

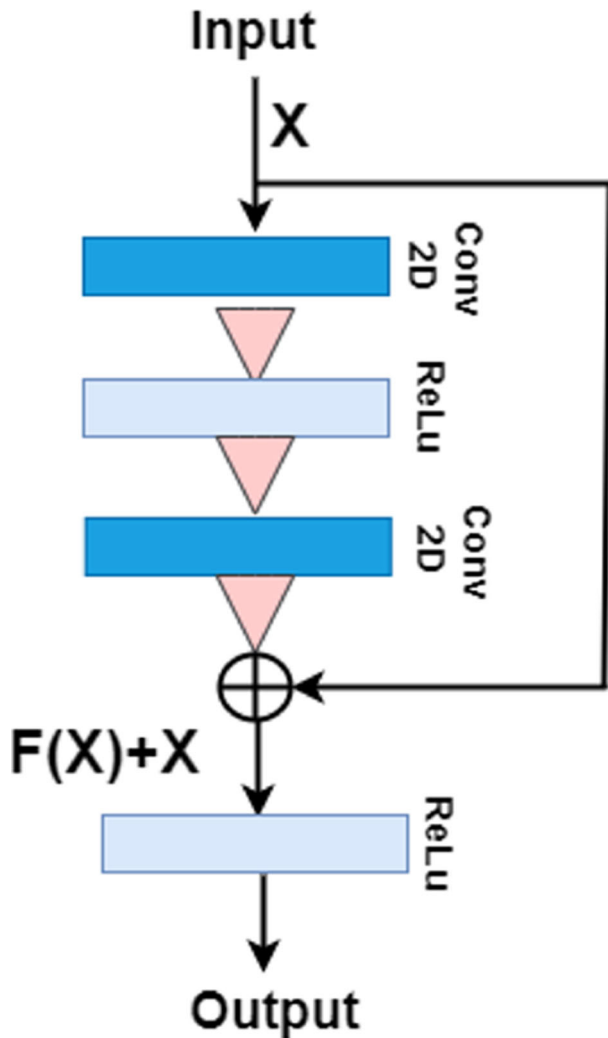


Fig. 3 Residual block used in LiverNet 2.x architecture

the trainable parameters of a few convolutional ($\otimes$) fully connected layers (FC), standard non-trainable functions such as *AveragePool* and *MaxPool* (which implements a global average and max pooling here) and the sigmoid function (σ). The functions f_c and f_s are described in (4), and (5).

$$f_c(F) = \sigma(FC(AveragePool(F)) + FC(MaxPool(F))) \quad (4)$$

$$f_s(F) = \sigma(w \otimes [f_{avg}(F); f_{max}(F)]) \quad (5)$$

where w is one the trainable convolutional kernels used in the convolutional layer and $f_{avg} : \mathbb{R}^{C \times H \times W} \rightarrow \mathbb{R}^{1 \times H \times W}$ and $f_{max} : \mathbb{R}^{C \times H \times W} \rightarrow \mathbb{R}^{1 \times H \times W}$ takes the average and maximum of values across the channels respectively. The ';' refers to channel-wise concatenation.

4. DenseASPP block Unlike the atrous spatial pyramid pooling (ASPP) block used in LiverNet, LiverNet 2.0 uses a modified form of ASPP block, known as DenseASPP. The ASPP block is responsible for increasing the receptive field of the convolution operation using stacked dilated convolutional layers [11]. The increased receptive field helps the network generalize the global patterns in the input feature map. The mathematical expression of dilated convolution is expressed in (6).

$$y[i][j] = \sum_m \sum_n x[i + r \cdot m][j + r \cdot n] \cdot w[m][n] = x \otimes w_r \quad (6)$$

where, $y[i][j]$ is the resultant tensor at the spatial location (i, j) , x is the tensor volume and w_r is a convolution filter. The dilation rate r , increases the receptive area of each convolution filter. Using these dilated convolutions of different kernel sizes increases the overall receptive area of the ASPP Block.

According to DeepLabV3 [10], there are two modes, cascading and parallel connection of atrous convolutional layers. The ASPP block in the LiverNet is the parallel mode which generates features with different receptive field scales. The DenseASPP, which we have adopted, involves stacking up dilated layers and connecting them with dense-like connections. This DenseASPP is motivated by the DenseNet-121 [20], discussed among the state-of-the-art architectures. This modification not only has a larger receptive field of features but also has a denser feature pyramid. It means it covers features at various scales as well as there is the involvement of more pixels in convolution. All the atrous convolution outputs are densely connected, such that the layers with small dilation rates are fed into the following layers (with higher dilation rates). The pooling outputs as well as 1×1 convolution outputs are directly concatenated with the atrous outputs. The final concatenated output is then passed through a 1×1 convolution layer to maintain the same number of channels as that of the input. Each convolution layer in this ASPP block is followed by a batch normalization and ReLU activation function. The output of this block is then sent to the bilinear upsampling layer. To decrease the total number of trainable parameters, the DenseASPP block was implemented using separable convolutional [14] rather than standard convolution. The DenseASPP block we used in LiverNet 2.0 is similar to [45], the only difference being the inclusion of 1×1 convolution and global pooling layers as well. Apart from 1×1 convolution and global pooling operations, all the 3×3 convolutions with dilation rates of 2, 3, 4, and 6 are densely connected as shown in Fig. 4.

3.2 LiverNet 2.1

The architecture, called LiverNet 2.1 is different from LiverNet 2.0 only in the modification of the ASPP block, which is discussed below. The main purpose of using an ASPP block is to increase the receptive field of the convolutions. But due to the channel-wise concatenation of the outputs of several blocks with different dilation rates, the channel dimension of the final output increase drastically. Instead, we planned to use fewer concatenations, while maintaining a large receptive field by stacking the dilated convolutional blocks in a tree-like fashion. According to [45], stacking a convolutional layer with a lower dilation rate below a block with a higher dilation rate increases the sampling rate of filtering. This leads to a larger receptive field, as well as a better sampling of the input feature maps. The convolutional layers are arranged to form the nodes of a binary-tree-like structure, as shown in Fig. 5, with the input feature map coming from the root and the dilation rate increasing as the tree grows. The highest dilation rate near the leaves increases the receptive field to a large

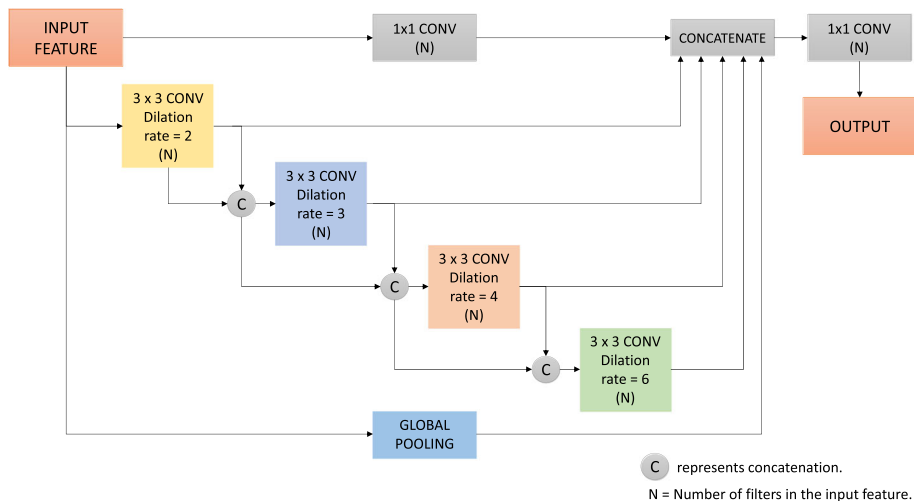


Fig. 4 DenseASPP block used in LiverNet 2.0

extent, whereas the blocks with a lower dilation rate near the root increases the density of the sampling rate.

The functions $f_0, f_1 : \mathbb{R}^{C_{in} \times H \times W} \rightarrow \mathbb{R}^{C_0 \times H \times W}$, $f_{00}, f_{01}, f_{10}, f_{11} : \mathbb{R}^{C_0 \times H \times W} \rightarrow \mathbb{R}^{C_1 \times H \times W}$ are convolutional blocks with their filters having increased receptive fields, with C_{in} being the number of input channels and C_0, C_1 chosen appropriately.

$$y_0 = f_0(X_{in}) = X_{in} \otimes w_{r_0} \quad (7)$$

$$y_1 = f_1(X_{in}) = X_{in} \otimes w_{r_1} \quad (8)$$

$$y_{00} = f_{00}(y_0) = y_0 \otimes w_{r_{00}} \quad (9)$$

$$y_{01} = f_{01}(y_0) = y_0 \otimes w_{r_{01}} \quad (10)$$

$$y_{10} = f_{10}(y_1) = y_1 \otimes w_{r_{10}} \quad (11)$$

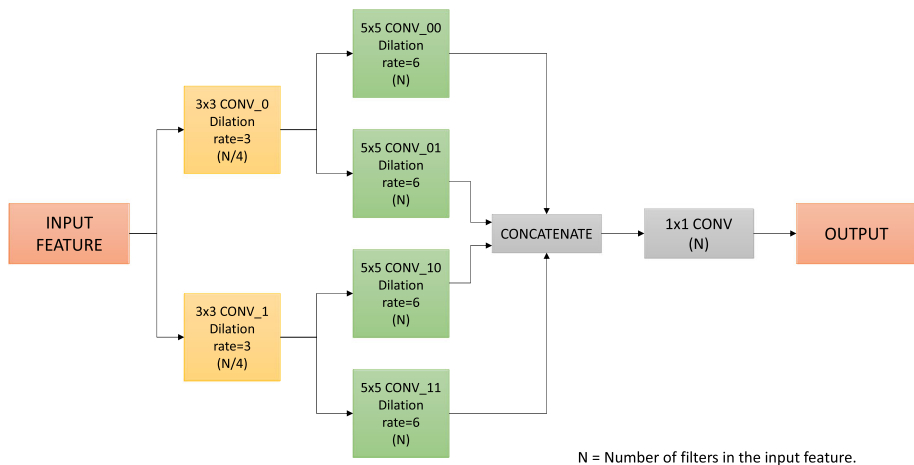


Fig. 5 Modified ASPP block in LiverNet 2.1

$$y_{11} = f_{11}(y_1) = y_1 \otimes w_{r_{11}} \quad (12)$$

$$y_{concat} = [y_{00}; y_{01}; y_{10}; y_{11}] \quad (13)$$

$$X_{out} = y_{concat} \otimes w_p \quad (14)$$

where $r_0, r_1, r_{00}, r_{01}, r_{10}, r_{11}$ are the dilation rates of each convolutions, ‘;’ denotes channel-wise concatenations and w_p is a 1×1 convolution filter. The number of filters w_p are chosen such that the output has the same number of channels as the input X . For this work, we have used $r_0 = r_1 = 3, r_{00} = r_{01} = r_{10} = r_{11} = 6, C_0 = C_{in}/4$ and $C_1 = C_{in}$. y_0 and y_1 are the first two branches of tree, while $y_{00}, y_{01}, y_{10}, y_{11}$ are the branches at next label. Equations (7) to (14) represents the operation from input to output of the modified ASPP block is used in LiverNet 2.1.

LiverNet 2.0 leverages the strength of dense connection as well as the effect of convolution using different dilation rates. LiverNet 2.1 used a stack of dilated convolution blocks in a tree-like fashion, where the aim is to reduce the complexity. Both forms of LiverNet 2.X receives collective information from all preceding layers and passed it to the subsequent layers.

4 Implementation and training

4.1 Datasets

Two datasets, namely, KMC liver and TCGA liver were used for the training and testing of the model. Both datasets are initially used in [1]. The detail about data acquisition and preparation of KMC liver and TCGA liver datasets are given in [1]. The overview of KMC and TCGA dataset splits is given in Tables 2 and 3. The Visualization of different grades of Hepatocellular carcinoma (HCC) tissues is shown in Fig. 6. Grade-0 Cells are well arranged and are normal in number. A normal glomerulus structure was seen in Grade-0, Grade-1 nucleoli are not visible even at 400X magnification whereas Grade-2 nucleoli are seen as eosinophilic at 400X magnification but not very prominent at 100X magnification. A slightly irregular contour compared to normal nuclei can be observed in the case of Grade-2. Grade-3 pronounced nuclear pleomorphism, and nuclei have a more irregular contour compared to normal nuclei. Hepatocellular carcinoma is the most common form of liver cancer. The Cancer Genome Atlas-liver hepatocellular carcinoma (TCGA-LIHC) dataset, which included 334 HCC samples. Three different classes of TCGA liver histopathology images are shown in Fig. 7.

4.2 Loss function

For experimentation of deep learning models, we have used the categorical cross entropy (CCE) loss [8, 41] as the criterion for the gradient descent optimization. The feed-forward operation of any convolutional neural network can be represented as function $f(x, w) : \mathbb{R}^{C_{in} \times H \times W} \longrightarrow \mathbb{R}^C$, where x is the input to the network, w as the weights of the network and C as the number of classes for cancer grade classification. The ground truth label y is a one-hot encoded vector encoding the class information. The output of the network to an input x_{in} is given by,

$$\hat{y} = f(x_{in}, w)$$

Table 2 KMC Dataset splits for different approaches

Approaches	Divisions	Grade 0	Grade 1	Grade 2	Grade 3
End-to-end training	Training	672	640	670	665
	Validation	70	80	80	50
	Testing	70	80	80	50
Cross Validation	Fold 1	70	64	68	70
	Fold 2	74	64	68	70
	Fold 3	70	64	68	70
	Fold 4	70	64	64	70
	Fold 5	70	64	64	56

Table 3 TCGA Dataset splits for different approaches

Approaches	Divisions	Normal	Primary	Secondary
End-to-end training	Training	200	240	124
	Validation	25	30	15
	Testing	25	30	16
Cross Validation	Fold 1	50	60	35
	Fold 2	50	60	30
	Fold 3	50	60	30
	Fold 4	50	60	30
	Fold 5	50	60	30

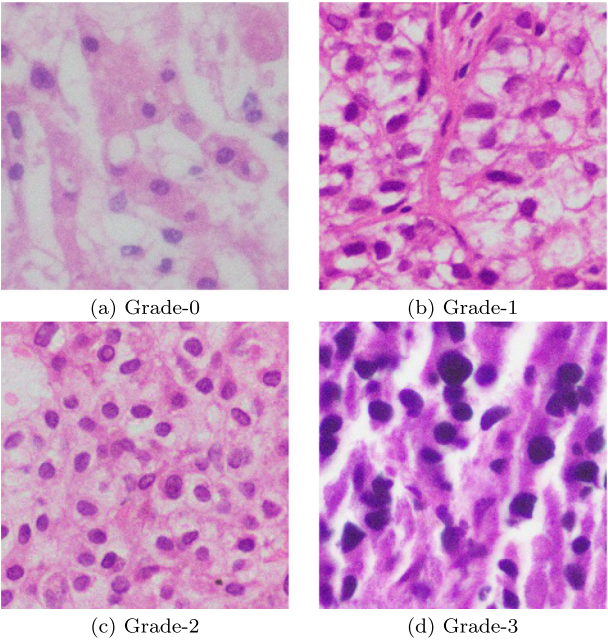


Fig. 6 Visualization of different grades of Hepatocellular carcinoma (HCC) tissues

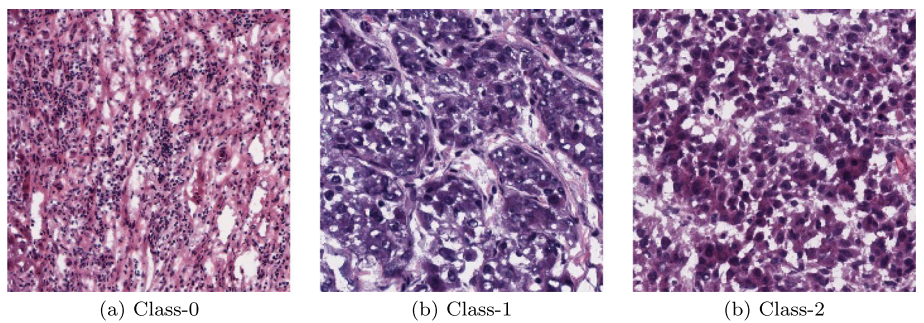


Fig. 7 TCGA histopathology images for each class: Normal, Primary tumor, Recurrent (Secondary) tumor from left to right

The CCE loss, to measure the dissimilarity between y and $\hat{y}$ is given by,

$$CCE = - \sum_{i=1}^C y_i \cdot \log(\hat{y}_i)$$

4.3 Network training

The complete implementation was done in Python 3.8 using PyTorch 1.9.0 and Google Colab standard GPU as computational hardware. For all layers in the network, the learnable parameters (weights) were initialized with the *kaiming-uniform* initialization, also known as He-Initialization [17]. Categorical cross-entropy loss is used as a loss function. All these models were trained with a batch size of 4 for 40 epochs, using the Adam optimizer [23]. In addition, dropouts [6] were also adopted as a regularization technique [25] for the parameters. Multiple trials on sets of hyper-parameters have been performed. The final selection of hyper-parameters is based on maximum accuracy among all variations. Important deep learning tools and hyper-parameters are shown in Table 4.

We employed two approaches for the training procedure. The first one being an end-to-end approach, in which the dataset was split into train, test and validation folds, with a division of 80% for training and 10% each for validation as well as testing. The loss and accuracy plots for LiverNet 2.0 on KMC and TCGA datasets are illustrated in Figs. 8 and 9

Table 4 Deep learning tools and details of hyperparameters

Software used	Python 3.8 using PyTorch 1.9.0
Computational hardware	Google Colab standard GPU
Loss function	Categorical cross-entropy
Optimizer	Adam (Initial learning rate = 0.0001)
Activation function	ReLu and Softmax
Parameter initialization	Kaiming-uniform initialization
Learning rate scheduler	Factor=0.5, patience=5, min-delta=0.0001, Monitor-Validation acc.
Batch size	4
Convolution filter	32, 64
Training epochs	40

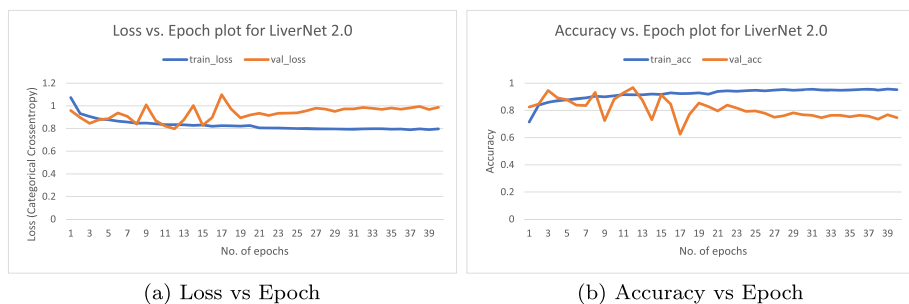


Fig. 8 Plots of Loss and Accuracy for LiverNet 2.0 on KMC dataset

respectively. The loss and accuracy plots for LiverNet 2.1 on KMC and TCGA datasets are illustrated in Figs. 10 and 11 respectively.

The second approach we adopted was a K -fold cross-validation. This approach involves the division of the dataset into K folds. The model was trained and tested K times by taking K^{th} fold for testing and the remaining $K - 1$ folds for training. In this paper, we selected $K = 5$. The results for the cross-validation were calculated by taking the averages of each fold as the test set. This approach ensures that the trained models are in no way biased toward any particular set of examples.

5 Ablation study

The proposed LiverNet is based on the atrous spatial pyramid pooling (ASPP) block, residual block, fully connected layers (Linear block), and attention block. A number of different configuration has been trained and the contributions of each individual block are verified. The proposed model effectively utilizes the advantages of residual connection, ASPP block, Linear block, and Attention block. The effectiveness of each block is evaluated by detaching them from the proposed model and the obtained results are given in Table 5.

These variations are given as CNN-1, CNN-2, CNN-3, CNN-4, and finally, the proposed model, where all blocks are jointly worked. Each variation leads to a reduction in the results and the proposed model, where contributions of all these blocks are involved, is giving the best result than the variations. The CNN-1 results in a reduction of 4.83% in Precision, 7.58% in Recall, 6.69% in dice coefficient, and 7.58% in Accuracy from the proposed model

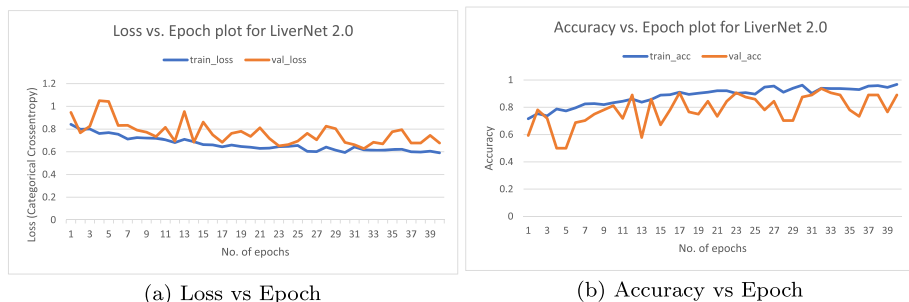


Fig. 9 Plots of Loss and Accuracy for LiverNet 2.0 on TCGA dataset

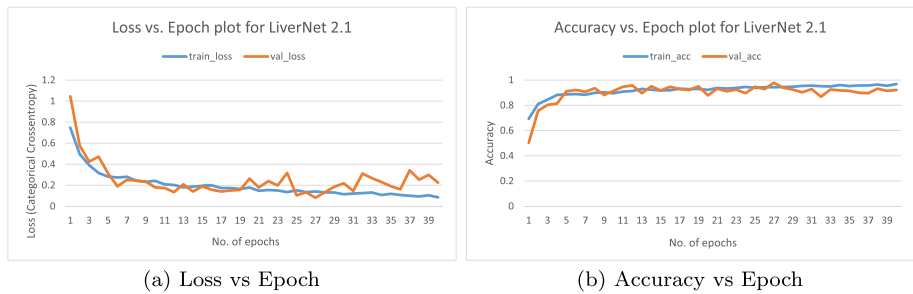


Fig. 10 Plots of Loss and Accuracy for LiverNet 2.1 on KMC dataset

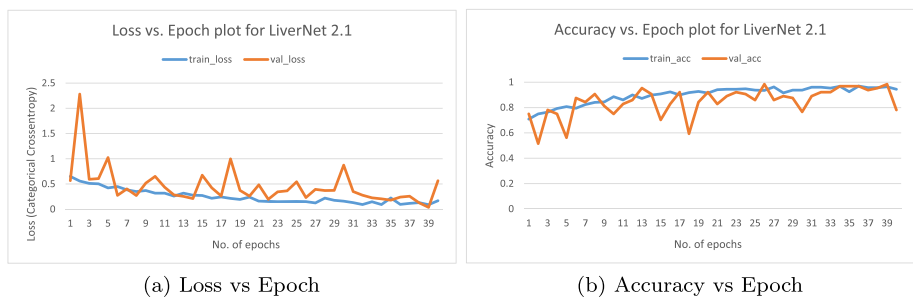


Fig. 11 Plots of Loss and Accuracy for LiverNet 2.1 on TCGA dataset

Table 5 Quality comparison of five variants of proposed LiverNet2.1 with KMC dataset

Model Variations	ASPP	Residual block	Linear block	Attention block	Precision	Recall	F1 Score	Accuracy
CNN-1	×	✓	✓	✓	0.9237	0.8956	0.9045	0.9
CNN-2	✓	×	✓	✓	0.9248	0.9112	0.9144	0.9071
CNN-3	✓	✓	×	✓	0.9147	0.9044	0.9078	0.9035
CNN-4	✓	✓	✓	×	0.9169	0.9131	0.9131	0.9071
Proposed	✓	✓	✓	✓	0.9740	0.9714	0.9714	0.9714

Table 6 Quality comparison of five variants of proposed LiverNet2.1 with TCGA dataset

Model Variations	ASPP	Residual block	Linear block	Attention block	Precision	Recall	F1 Score	Accuracy
CNN-1	×	✓	✓	✓	0.9227	0.9365	0.9274	0.931
CNN-2	✓	×	✓	✓	0.9203	0.9389	0.9211	0.9241
CNN-3	✓	✓	×	✓	0.9661	0.9598	0.9625	0.9589
CNN-4	✓	✓	✓	×	0.9369	0.9515	0.9415	0.9448
Proposed	✓	✓	✓	✓	0.9720	0.9714	0.9713	0.9714

for the KMC dataset. Among all the variations, the least value of F1-Score for the KMC dataset by the variation named CNN-3. The obtained results of different CNN variations for the TCGA dataset are given in Table 6. The reported results in Tables 5 and 6 shows that the contributions of individual block can not be ignored to produce the final model.

6 Results and discussions

6.1 Performance evaluation metrics

Performance evaluation metrics [1, 26, 29] are the metrics used to quantify the similarity between the predicted output of the model and the ground-truth label. The metrics used in this study are as follows:

$$\begin{aligned} Precision &= \frac{TP}{TP + FP} & Recall &= \frac{TP}{TP + FN} \\ F1\ score &= \frac{2TP}{2TP + FN + FP} \\ Accuracy &= \frac{TP + TN}{TP + FP + TN + FN} \end{aligned}$$

TP : True Positives, FP : False Positives, FN : False Negatives, TN : True Negatives

We used 'weighted average' for the calculation of the metrics such as Precision, Recall, and F1 score to account for class imbalance in the dataset.

6.2 Experimental results

The training of proposed LiverNet 2.x and other existing deep learning models were carried out by using an end-to-end (train-test-validation split) manner as well as using 5-fold cross-validation manner. The results of proposed LiverNet 2.x and other existing deep learning models on test KMC dataset are given in Tables 7 and 9, and on the test TCGA dataset are given in Tables 8 and 10. The 4-class confusion matrices obtained by training the LiverNet 2.x on KMC dataset are shown in Fig. 12 and the 3-class confusion matrices for the same on TCGA dataset are shown in Fig. 13. The matrices illustrate the classification performance of the models on the corresponding datasets. They depict the number of examples correctly and incorrectly classified to various classes. By observing the confusion matrices, it can be inferred that most of the examples in the testing set have been correctly classified to their respective classes. But to get a quantitative measure of the performances of each model, we refer to the Tables 7, 8, 9 and 10. The loss-vs-epoch (validation loss) and the accuracy-vs-epoch (validation accuracy) plots obtained during the end-to-end training of the models are shown in Fig. 8, 9, 10 and 11. These plots illustrate how the loss and the accuracy values improved over 40 epochs of training.

6.3 Discussion

The performance of the LiverNet 2.x have been compared with ResNet (2016) [18], DenseNet (2017) [20], BreastNet (2020) [38], LiverNet (2021) [1], Hameed et al. [15], and

Table 7 Performance comparison of the proposed architectures with competitive architectures on test KMC dataset using end-to-end training approach

Metrics	Class	He et al. [18]	Huang et al. [20]	Togacar et al. [38]	Aatresh et al. [1]	Abbasniya et al. [2]	Hameed et al. [15]	LiverNet 2.0	LiverNet 2.1
Precision	Grade-0	0.9241	1.0000	1.0000	1.0000	1.000	1.0000	0.9859	1.0000
	Grade-1	0.8378	0.8422	0.8372	0.8587	0.8865	0.8953	0.9620	1.0000
	Grade-2	0.8438	0.8501	0.8890	0.9853	0.8753	0.9342	0.9625	0.9091
	Grade-3	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	Average	0.9014	0.9230	0.9335	0.9554	0.9404	0.9573	0.9749	0.9740
Recall	Grade-0	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	Grade-1	0.8708	0.9250	0.9333	0.9875	0.9038	0.9625	0.9500	0.9000
	Grade-2	0.7833	0.8375	0.7908	0.8375	0.8780	0.8875	0.9625	1.0000
	Grade-3	0.9000	0.8600	0.8600	1.0000	0.9335	0.9600	1.0000	1.0000
	Average	0.8885	0.9056	0.9187	0.9500	0.9288	0.9525	0.9750	0.9714
F1 score	Grade-0	0.9875	1.0000	1.0000	1.0000	1.0000	1.0000	0.9929	1.0000
	Grade-1	0.8515	0.8808	0.8791	0.9186	0.8990	0.9277	0.9560	0.9474
	Grade-2	0.8091	0.8427	0.8417	0.9054	0.8740	0.9102	0.9625	0.9524
	Grade-3	0.9463	0.9240	0.9247	1.0000	0.9674	0.9795	1.0000	1.0000
	Average	0.8911	0.9118	0.9213	0.9497	0.9341	0.9543	0.9749	0.9714
Accuracy	-	0.8832	0.9070	0.9142	0.9500	0.9251	0.9500	0.9750	0.9714
Parameters	-	23.59 M	7.04 M	0.60 M	0.573 M	54.341 M	2.236 M	0.37 M	0.273 M

Table 8 Performance comparison of the proposed architectures with competitive architectures on test TCGA dataset using end-to-end training approach

Metrics	Class	He et al. [18]	Huang et al. [20]	Togacar et al. [38]	Aatresh et al. [1]	Abbasniya et al. [2]	Hameed et al. [15]	LiverNet 2.0	LiverNet 2.1
Precision	Class-0	0.9897	0.9225	0.9958	0.9615	0.9804	0.8906	0.9600	0.9600
	Class-1	0.9911	0.9150	0.9572	0.9643	0.9649	0.8888	0.9063	0.9667
	Class-2	0.8427	0.9022	0.9215	0.8824	0.8918	0.6888	1.0000	1.0000
	Average	0.9405	0.9348	0.9582	0.9449	0.9457	0.8228	0.9463	0.9720
Recall	Class-0	1.0000	0.9166	1.0000	1.0000	1.0000	0.9800	0.9600	1.0000
	Class-1	0.8416	0.8791	0.9333	0.9000	0.9166	0.6660	0.9667	0.9667
	Class-2	0.9892	0.9785	0.9462	0.9375	0.9428	0.8857	0.8750	0.9375
	Average	0.9436	0.9247	0.9598	0.9437	0.9531	0.8439	0.9437	0.9714
F1 score	Class-0	0.9937	0.9116	0.9979	0.9804	0.9900	0.9333	0.9600	0.9796
	Class-1	0.9090	0.9176	0.9450	0.9310	0.9401	0.7619	0.9355	0.9667
	Class-2	0.9092	0.9333	0.9336	0.9091	0.9166	0.7750	0.9333	0.9677
	Average	0.9373	0.9208	0.9588	0.9435	0.9489	0.8234	0.9436	0.9713
Accuracy	-	0.9399	0.9203	0.9609	0.9437	0.9517	0.8275	0.9437	0.9714
Parameters	-	23.59 M	7.04 M	0.60 M	0.573 M	54.341 M	2.236 M	0.37 M	0.273 M

Table 9 Performance comparison of the proposed architectures with competitive architectures on test KMC dataset using cross-validation approach

Metrics	Class	He et al. [18]	Huang et al. [20]	Togacar et al. [38]	Aatresh et al. [1]	Abbasniya et al. [2]	Hameed et al. [15]	LiverNet 2.0	LiverNet 2.1
Precision	Grade 0	0.9809	0.9759	0.9891	0.9944	0.9722	0.9896	0.9850	0.9919
	Grade 1	0.8255	0.8129	0.8496	0.8601	0.8860	0.9216	0.9299	0.8372
	Grade 2	0.7202	0.7534	0.7394	0.7945	0.8068	0.8227	0.8366	0.8554
	Grade 3	0.9964	0.9809	0.9898	0.9972	1.0000	1.000	0.9856	0.9830
	Average	0.8832	0.8832	0.8942	0.9136	0.9162	0.9334	0.9355	0.9171
Recall	Grade 0	1.0000	0.9943	1.0000	1.0000	1.0000	1.0000	0.9971	1.0000
	Grade 1	0.7094	0.8156	0.7750	0.8250	0.875	0.9169	0.7969	0.8625
	Grade 2	0.8658	0.8096	0.8616	0.8638	0.8875	0.9099	0.9226	0.7952
	Grade 3	0.8771	0.8371	0.8914	0.9400	0.8200	0.8041	0.9857	0.9800
	Average	0.8667	0.8666	0.8849	0.9096	0.8956	0.9077	0.9285	0.9083
F1 score	Grade 0	0.9903	0.9846	0.9945	0.9972	0.9860	0.9987	0.9908	0.9959
	Grade 1	0.7536	0.8124	0.8091	0.8405	0.8805	0.9313	0.8485	0.8418
	Grade 2	0.7783	0.7680	0.7942	0.8265	0.8452	0.8652	0.8739	0.8021
	Grade 3	0.9246	0.8820	0.9323	0.9633	0.9010	0.9167	0.9856	0.9811
	Average	0.8650	0.8643	0.8852	0.9091	0.9031	0.9279	0.9271	0.9062
Accuracy	-	0.8652	0.8651	0.8841	0.9093	0.9000	0.9134	0.9285	0.9114
Parameters	-	23.59 M	7.04 M	0.60 M	0.573 M	54.341 M	2.236 M	0.37 M	0.273 M

Table 10 Performance comparison of the proposed architectures with competitive architectures on test TCGA dataset using cross-validation approach

Metrics	Class	He et al. [18]	Huang et al. [20]	Togacar et al. [38]	Aatresh et al. [1]	Abbasniya et al. [2]	Hameed et al. [15]	LiverNet 2.0	LiverNet 2.1
Precision	Class 0	0.9516	0.9510	0.9884	0.9923	0.9804	0.8940	0.9612	0.9923
	Class 1	0.9266	0.9686	0.9695	0.9965	0.8615	0.8171	0.9624	0.9902
	Class 2	0.8873	0.8902	0.9081	0.9239	0.8965	0.6210	0.9056	0.9410
	Average	0.9268	0.9451	0.9627	0.9790	0.9128	0.7773	0.9496	0.9795
Recall	Class 0	0.9960	0.9840	1.0000	1.0000	1.0000	0.9066	0.9839	0.9960
	Class 1	0.8900	0.8700	0.9367	0.9500	0.9333	0.5788	0.9133	0.9567
	Class 2	0.8552	0.9610	0.9438	0.9933	0.7428	0.8686	0.9543	0.9867
	Average	0.9199	0.9304	0.9607	0.9773	0.8920	0.7846	0.9474	0.9774
F1 score	Class 0	0.9730	0.9647	0.9941	0.9961	0.9900	0.8896	0.9722	0.9941
	Class 1	0.9063	0.9091	0.9521	0.9723	0.8960	0.6561	0.9367	0.9724
	Class 2	0.8548	0.9177	0.9234	0.9567	0.8125	0.7220	0.9277	0.9620
	Average	0.9186	0.9307	0.9607	0.9773	0.8995	0.7559	0.9474	0.9775
Accuracy	-	0.9205	0.9308	0.9604	0.9772	0.9103	0.7709	0.9474	0.9774
Parameters	-	23.59 M	7.04 M	0.60 M	0.573 M	54.341 M	2.236 M	0.37 M	0.273 M

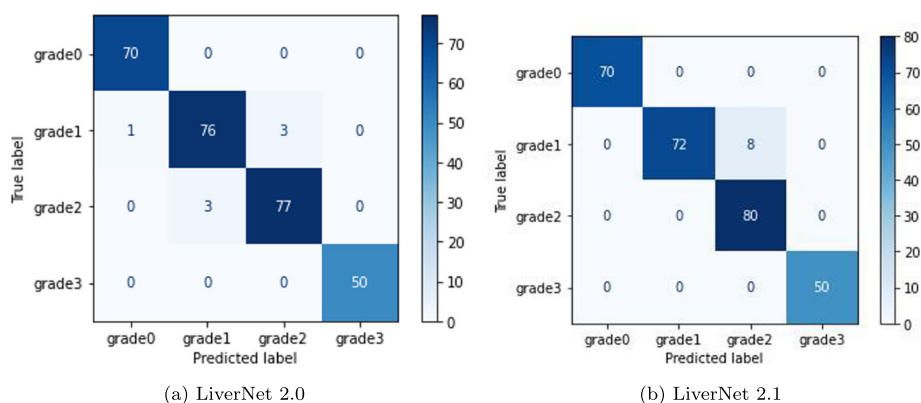


Fig. 12 Confusion matrices on KMC dataset

Abbasniya et al. [2]. For the KMC dataset, the proposed LiverNet2.0 achieved 97.50% accuracy, which is the highest for four class problems. In this case, ResNet architecture achieves 88.32% accuracy which is the minimum among all comparison models. The resNet model has 23 Million trainable parameters which are approximately 86 times that of the proposed model. The architecture proposed by Hameed et al. [15] has maximum trainable parameters. LiverNet2.1 achieved 97.14% accuracy, 97.40% precision, 97.14% recall, and 97.14% F1-score value. For the TCGA dataset, the proposed LiverNet2.0 achieved 97.74% accuracy, 94.96% precision, 94.74% recall, and 94.74% F1-score value. In this case, the architecture proposed by Hameed et al. [15] achieve 77.09% accuracy which is the minimum among all comparison model. This architecture also imposed 54.34 Million trainable parameters which is approximately 199 times than the proposed model. LiverNet2.1 achieved 97.74% accuracy, 97.95% precision, 97.74% recall, and 97.75% F1-score value for three class problems. It can be observed that the proposed models not only outperform the other competitive models w.r.t. their performance metrics but also with respect to their computational complexities, as depicted in Fig. 15. LiverNet 2.x achieves better performance with a lesser number of trainable parameters, with its variant LiverNet 2.1 achieves it with the lowest

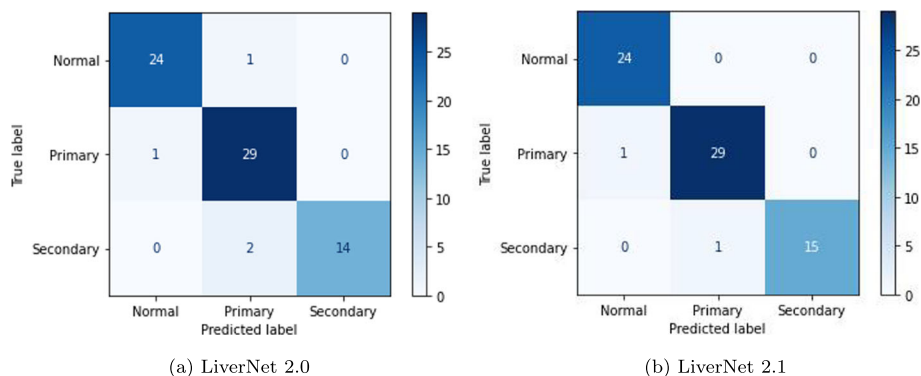


Fig. 13 Confusion matrices on TCGA dataset

number of parameters. Figure 14 shows that LiverNet 2.0 performs slightly better than LiverNet 2.1 on the KMC dataset. But on TCGA test data, LiverNet 2.1 achieves a significantly better performance than LiverNet 2.0. This behavior can be justified by the lesser computational complexity of LiverNet 2.1, which is able to avoid overfitting on the TCGA dataset with a smaller size. These contrasting behaviors can be explained as a trade-off between the computational complexity, dataset size, and the resulting performance of the models. So we can infer that given a dataset of considerable size, LiverNet 2.0 can learn more complex features as compared to LiverNet 2.1. The architecture LiverNet 2.1 achieves considerably better performance on small datasets, which is generally what the data scientists and researchers encounter in medical data analysis. Table 11 summarizes the computational complexity analysis, including the total number of trainable parameters and floating-point operations (FLOPs). Table 11 shows a modified form of ASPP blocks is effective in feature extraction and using fewer concatenations layers while maintaining a sizeable receptive field is attributed to better accuracy. Combining the usage of features at various scales with multiple receptive areas was the key idea of the proposed work. Our study is different from the existing study due to the following reasons. (a) The final goal of using Artificial Intelligence (AI) in healthcare is to design a fast and precise cancer detection system. (b) The relevance and potential of an automatic framework for the classification of histopathology images have been demonstrated by many authors, but the reported result is still sub-optimal for clinical use. (c) The proposed method provides a computationally efficient and fully automated deep learning framework, which is one step ahead compared to existing solutions. (d) Our approach is generalized and effectively works on two different histopathology datasets. (e) It reduces the need for extensive computational complexity (Fig. 15).

6.4 Limitations and future work

Reported results of two different datasets show that there is a lack of generalizability in the classification of liver tissues. Though we have made use of two adequate and reliable datasets, one of the major challenges in this study was the limited availability of histopathology data for higher-graded cancer types. Since most of the patients belonging to each subtype is a highly non-uniform distribution, this challenge cannot be completely overcome. But we definitely hope to find larger and better training data for future studies in this domain, leading to much better results by the automation process, which is our ultimate goal.

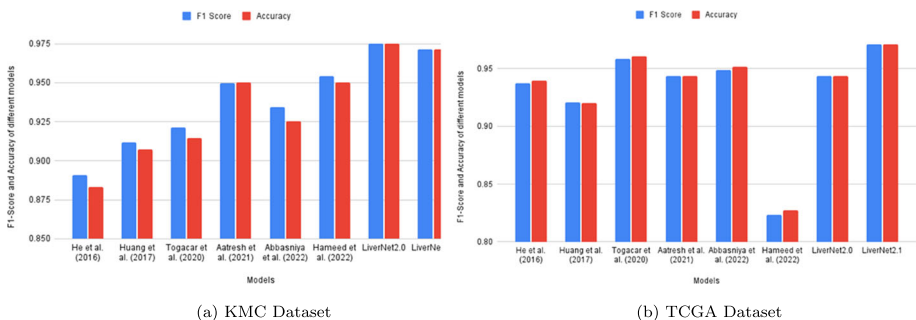


Fig. 14 Comparison plots for F1 score and accuracy

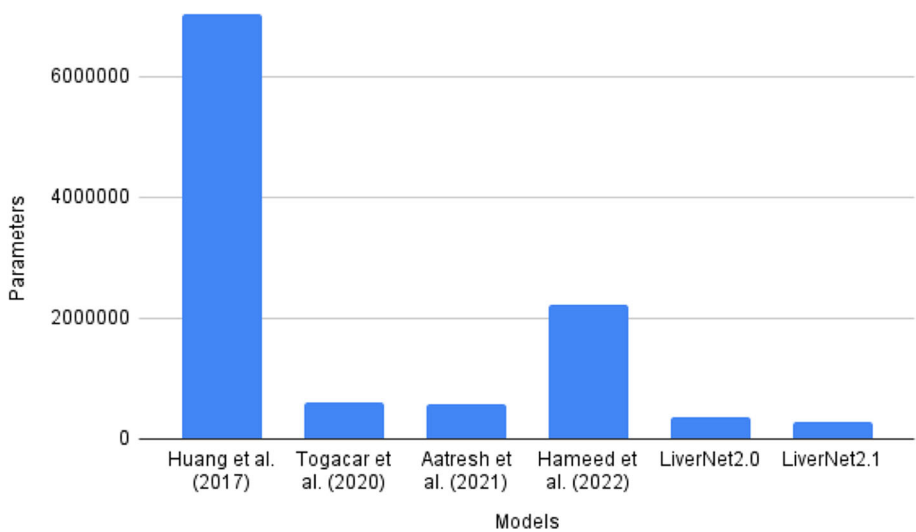


Fig. 15 Comparison of number of parameters of the architectures

7 Conclusion

This paper proposes LiverNet 2.0 and LiverNet 2.1 architectures which give superior performance compared to its predecessor, the LiverNet, as well as the state-of-the-art deep learning models. In this study, we presented two different feature extraction CNN blocks and their performance on the two liver HCC histopathology datasets. The DenseASPP and modified ASPP of LiverNet 2.x architecture address two major problems. Classifying different subtypes and grades from complex structured histopathology images with different histology and microscopic characteristics was the first major concern. This issue is solved by the effective use of the dense connection, residual connection, and dilated convolution in implemented CNN blocks. The second major concern was to reduce the computational complexity and total trainable parameters, which was effectively handled by incorporating depth-wise separable convolution in ASPP block. Our experimental results showed that LiverNet 2.0 gives an accuracy of 0.9750 and F1-score of 0.9749 on the KMC dataset, whereas

Table 11 Computational complexity analysis of different models

Model	Parameters (M = 10 ⁶)	FLOPs (G = 10 ⁹)
He et al. [18]	23.60 M	7.75 G
Huang et al. [20]	7.04 M	5.70 G
Togacar et al. [38]	0.606 M	5.68 G
Aatresh et al. [1]	0.574 M	3.72 G
Abbasniya et al. [2]	54.34 M	13 G
Hameed et al. [15]	2.23 M	5.61 G
Abbasniya et al. [2]	54.34 M	13 G
LiverNet 2.0	0.37 M	2.79G
LiverNet 2.1	0.273 M	2.71G

the LiverNet 2.1, with nearly half as the number of parameters as LiverNet, gives an accuracy of 0.9714 and F1-score of 0.9714 on the KMC dataset. We also performed a 5-fold cross-validation to check any overfitting issues. The reported results show a similar trend in terms of accuracy-vs-model complexity.

Acknowledgements The authors would like to thank the handing editor, editor-in-chief and anonymous reviewers for their valuable suggestions and comments which helped to improve the quality of the research paper.

Data availability The datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

Declarations

Conflict of interests Authors declare that they have no conflict of interest.

References

1. Aatresh AA, Alabhya K, Lal S, Kini J, Saxena PUP (2021) Livernet: efficient and robust deep learning model for automatic diagnosis of sub-types of liver hepatocellular carcinoma cancer from H&E stained liver histopathology images. *Int J Comput Assist Radiol Surg* 16(9):1549–1563. <https://doi.org/10.1007/s11548-021-02410-4>
2. Abbasniya MR, Sheikholeslamzadeh SA, Nasiri H, Emami S (2022) Classification of breast tumors based on histopathology images using deep features and ensemble of gradient boosting methods. *Comput Electr Eng* 103:108382
3. Afza F, Sharif M, Khan MA, Tariq U, Yong H-S, Cha J (2022) Multiclass skin lesion classification using hybrid deep features selection and extreme learning machine. *Sensors* 22(3):799. <https://doi.org/10.3390/s22030799>
4. Alhindi TJ, Kalra S, Ng KH, Afrin A, Tizhoosh HR (2018) Comparing LBP, HOG and Deep Features for Classification of Histopathology Images. 2018 International Joint Conference on Neural Networks (IJCNN), pp 1–7
5. Arooj S, Atta-Ur-Rahman, Zubair M, Khan MF, Alissa K, Khan MA, Mosavi A (2022) Breast cancer detection and classification empowered with transfer learning. *Front Public Health* 10:924432. <https://doi.org/10.3389/fpubh.2022.924432>
6. Cai S, Shu Y, Chen G, Ooi BC, Wang W, Zhang M (2019) Effective and efficient dropout for deep convolutional neural networks, *Computer vision and pattern recognition*, arXiv: [1904.03392](https://arxiv.org/abs/1904.03392)
7. Calderaro J, Ziol M, Paradis V, Zucman-Rossi J (2019) Molecular and histological correlations in liver cancer. *J Hepatol* 71(3):616–630
8. Chanchal AK, Kumar A, Lal S, Kini J (2021) Efficient and robust deep learning architecture for segmentation of kidney and breast histopathology images. *Comput Electric Eng* 92:107177. <https://doi.org/10.1016/j.compeleceng.2021.107177>
9. Chen C, Chen C, Ma M, Ma X, Lv X, Dong X, Yan Z, Zhu M, Chen J (2022) Classification of multi-differentiated liver cancer pathological images based on deep learning attention mechanism. *BMC Med Inform Decis Mak* 22(1):176. <https://doi.org/10.1186/s12911-022-01919-1>
10. Chen LC, Papandreou G, Kokkinos I, Murphy K, Yuille AL (2018) Deeplab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs. *IEEE Trans Pattern Anal Mach Intell* 40(4):834–848. <https://doi.org/10.1109/TPAMI.2017.2699184>
11. Chen LC, Papandreou G, Schroff F, Adam H (2017) Rethinking atrous convolution for semantic image segmentation. *Computer Vision and Pattern Recognition*. arXiv: [1706.05587v3](https://arxiv.org/abs/1706.05587v3) [cs.CV]
12. Chen M, Zhang B, Topatana W et al (2020) Classification and mutation prediction based on histopathology H&E images in liver cancer using deep learning. *NPJ Precis Onc* 4(14):1–7. <https://doi.org/10.1038/s41698-020-0120-3>
13. Chiang CH, Weng CL, Chiu HW (2021) Automatic classification of medical image modality and anatomical location using convolutional neural network. *PLoS One* 16(6):e0253205. <https://doi.org/10.1371/journal.pone.0253205>

14. Chollet F (2017) Xception: Deep learning with depthwise separable convolutions. In: 2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp 1800–1807, <https://doi.org/10.1109/CVPR.2017.195>
15. Hameed Z, Garcia-Zapirain B, Aguirre JJ (2022) Multiclass classification of breast cancer histopathology images using multilevel features of deep convolutional neural network. *Sci Rep* 12:15600. <https://doi.org/10.1038/s41598-022-19278-2>
16. Hao Y, Zhang L, Qiao S, Bai Y, Cheng R, Xue H, Hou Y, Zhang W, Zhang G (2022) Breast cancer histopathological images classification based on deep semantic features and gray level co-occurrence matrix. *PLOS ONE* 17(5):e0267955
17. He K, Zhang X, Ren S, Sun J (2015) Delving deep into rectifiers: Surpassing human-level performance on ImageNet classification. In: 2015 IEEE International Conference on Computer Vision (ICCV), pp 1026–1034, <https://doi.org/10.1109/ICCV.2015.123>
18. He K, Zhang X, Ren S, Sun J (2016) Deep Residual Learning for Image Recognition. In: 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Las Vegas, NV, pp 770–778, <https://doi.org/10.1109/CVPR.2016.90>
19. Houssein EH, Emam MM, Ali AA (2022) An optimized deep learning architecture for breast cancer diagnosis based on improved marine predators algorithm. *Neural Comput Applic* 34:18015–18033. <https://doi.org/10.1007/s00521-022-07445-5>
20. Huang G, Liu Z, Maaten LVD, Weinberger KQ (2017) Densely Connected Convolutional Networks. In: 2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp 2261–2269, <https://doi.org/10.1109/CVPR.2017.243>
21. Ioffe S, Szegedy C (2015) Batch normalization: Accelerating deep network training by reducing internal Covariate shift. *Machine Learning*. Arxiv: [1502.03167](https://arxiv.org/abs/1502.03167)
22. Kim T, Hedayat M, Vaitkus V, Belohlavek M, Krishnamurthy V, Borazjani I (2021) Automatic segmentation of the left ventricle in echocardiographic images using convolutional neural networks. *Quant Imaging Med Surg* 11(5):1763–1781
23. Kingma DP, Ba J (2015) Adam: A method for stochastic optimization, In international conference on learning representations, 9. arXiv: [1412.6980v9](https://arxiv.org/abs/1412.6980v9) [cs.LG]
24. Krizhevsky A, Sutskever I, Hinton GE (2012) ImageNet classification with deep convolutional neural networks. *Adv Neural Inf Process Syst* 25:1106–1114
25. Kukacka J, Golkov V, Cremers D (2017) Regularization for deep learning: A Taxonomy, Neural and evolutionary computing, [1710.10686](https://arxiv.org/abs/1710.10686)
26. Lal S, Das D, Alabhya K, Kanfode A, Kumar A, Kini J (2021) Nucleisegnet: Robust deep learning architecture for the nuclei segmentation of liver cancer histopathology images. *Comput Biol Med* 128:104075
27. Liu M, Hu L, Tang Y, Wang C, He Y, Zeng C, Lin K, He Z, Huo W (2022) A deep learning method for breast cancer classification in the pathology images. *IEEE J Biomed Health Inform* 26(10):5025–5032. <https://doi.org/10.1109/JBHI.2022.3187765>
28. Lu L, Daigle Jr BJ (2020) Prognostic analysis of histopathological images using pre-trained convolutional neural networks: application to hepatocellular carcinoma. *PeerJ* 12(8):e8668. <https://doi.org/10.7717/peerj.8668>
29. Naylor P, Lae M, Reyat F, Walter T (2019) Segmentation of nuclei in histopathology images by deep regression of the distance map. *IEEE Trans Med Imaging* 38(2):448–459
30. Nguyen TBT, Ngo MV, Nguyen VP (2022) Histopathological imaging classification of breast tissue for cancer diagnosis support using deep learning models. In: Vo NS, Vien QT, Ha DB (eds) *Industrial networks and intelligent systems*. INISCOM 2022. Lecture notes of the institute for computer sciences, social informatics and telecommunications engineering, vol 444. Springer, Cham
31. Othman E, Mahmoud M, Dhahri H, Abdulkader H, Mahmood A, Ibrahim M (2022) Automatic detection of liver cancer using hybrid pre-trained models. *Sensors (Basel)* 22(14):5429. <https://doi.org/10.3390/s22145429>
32. Puttagunta M, Ravi S (2021) Medical image analysis based on deep learning approach. *Multimed Tools Appl* 80:24365–24398. <https://doi.org/10.1007/s11042-021-10707-4>
33. Rehman MU et al (2022) A novel chaos-based privacy-preserving deep learning model for cancer diagnosis. *IEEE Trans Netw Sci Eng* 9(6):4322–4337. <https://doi.org/10.1109/TNSE.2022.3199235>
34. Srinidhi CL, Kim SW, Chen FD, Martel AL (2022) Self-supervised driven consistency training for annotation efficient histopathology image analysis. *Med Image Anal* 75:102256
35. Sun C, Xu A, Liu D, Xiong Z, Zhao F, Ding W (2020) Deep learning-based classification of liver cancer histopathology images using only global labels. *IEEE J Biomed Health Inform* 24(6):1643–1651. <https://doi.org/10.1109/JBHI.2019.2949837>

36. Szegedy C, Liu W, Jia Y, Sermanet P, Reed S, Anguelov D, Erhan D, Vanhoucke V, Rabinovich A (2015) Going deeper with convolutions. In: 2015 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp 1–9, <https://doi.org/10.1109/CVPR.2015.7298594>
37. Tabakov M, Zareba S, Podhorska Okolow M, Pula B (2013) Using fuzzy sugeno integral as an aggregation operator of ensemble of fuzzy decision trees in the recognition of HER2 breast cancer histopathology images. In: 2013 International Conference on Computer Medical Applications (ICCM), pp 1–6
38. Togacar M, Ozkurt KB, Ergen B, Comert Z (2020) Breastnet: A novel convolutional neural network model through histopathological images for the diagnosis of breast cancer. *Phys A: Stat Mech Appl* 545:123592
39. Travis WD, Brambilla E, Geisinger KR (2016) Histological grading in lung cancer: one system for all or separate systems for each histological type? *Eur Respir J* 47(3):720–3
40. Vu TH, Mousavi HS, Monga V, Rao UKA, Rao G (2015) DFDL: Discriminative feature-oriented dictionary learning for histopathological image classification. In: 2015 IEEE 12th International Symposium on Biomedical Imaging (ISBI) pp 99–994
41. Wei B, Han Z, He X, Yin Y (2017) Deep learning model based breast cancer histopathological image classification. In: 2017 IEEE 2nd International Conference on Cloud Computing and Big Data Analysis (ICCCBDA), pp 348–353, <https://doi.org/10.1109/ICCCBDA.2017.7951937>
42. Woo S, Park J, Lee JY, Kweon IS (2018) CBAM: Convolutional Block Attention Module. In: Ferrari V, Hebert M, Sminchisescu C, Weiss Y (eds) *Computer Vision – ECCV 2018*. ECCV 2018. Lecture Notes in Computer Science, vol 11211. Springer, Cham. https://doi.org/10.1007/978-3-030-01234-2_1
43. Xu X, Mao Y, Tang Y, Liu Y, Xue C, Yue Q, Liu Q, Wang J, Yin Y (2022) Classification of hepatocellular carcinoma and intrahepatic cholangiocarcinoma based on radiomic analysis. *Comput Math Methods Med* 2022:5334095. <https://doi.org/10.1155/2022/5334095>
44. Yang JD, Hainaut P, Gores GJ, Amadou A, Plymoth A, Roberts LR (2019) A global view of hepatocellular carcinoma: trends, risk, prevention and management. *Nat Rev Gastroenterol Hepatol* 16(10):589–604
45. Yang M, Yu K, Zhang C, Li Z, Yang K (2018) DenseASPP for Semantic Segmentation in Street Scenes. In: 2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp 3684–3692, <https://doi.org/10.1109/CVPR.2018.00388>

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

Affiliations

Amit Kumar Chanchal¹ · Shyam Lal¹  · Dipanshu Barnwal¹ · Prince Sinha¹ · Shrikant Arvvasu¹ · Jyoti Kini^{2,3}

Amit Kumar Chanchal
amit.chanchal01@gmail.com

Dipanshu Barnwal
dipanshu27dps@gmail.com

Prince Sinha
princesinha24august@gmail.com

Shrikant Arvvasu
ashrikant39@gmail.com

Jyoti Kini
kinijyoti@gmail.com

¹ Department of Electronics and Communication Engineering, National Institute of Technology Karnataka, Surathkal, Mangaluru 575025, Karnataka, India

² Department of Pathology, Kasturba Medical College, Mangalore, India

³ Department of Pathology, Manipal Academy of Higher Education, Manipal, India